



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
19-12-2021

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S(TEPS)]

Time : 3 Hours

PAPER-2

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

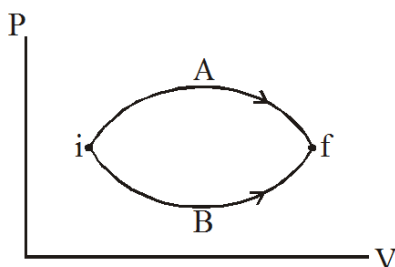
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. For the two processes shown. Which of the following is/are true :-



(A) $Q_A > Q_B$

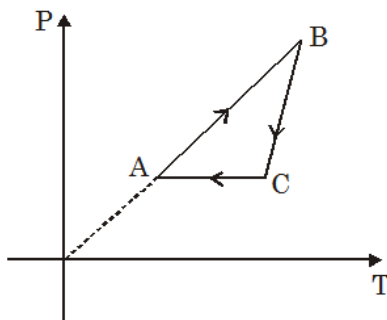
(B) $\Delta U_A = \Delta U_B$

(C) $Q_A < Q_B$

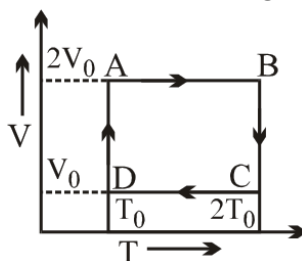
(D) $W_A > W_B$

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2. P-T graph for an ideal gas is shown in the figure. Choose the correct statement(s) :



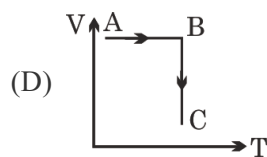
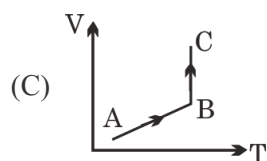
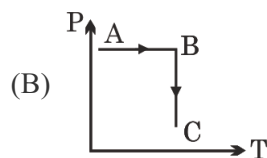
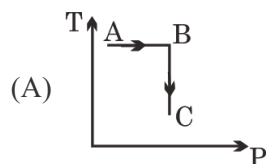
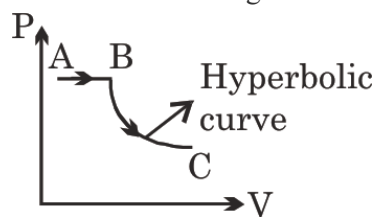
- (A) Process AB is isochoric
(B) Process BC is isothermal
(C) Process CA is isobaric
(D) $V_C > V_A$
3. One mole of an ideal gas is taken through the cyclic process shown in the V – T diagram where V = volume and T = absolute temperature of the gas. Which of the following statements are correct?



- (A) heat is given out by the gas
(B) heat is absorbed by the gas
(C) the magnitude of the work done by the gas is $RT_0 \ln 2$
(D) the magnitude of the work done by the gas is $V_0 T_0$.

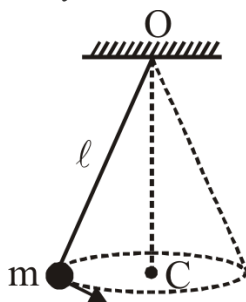
Space for Rough Work

4. A process is shown in the diagram. Which of the following curves may represent the same process?



Space for Rough Work

5. A small ball of mass m suspended from the ceiling at a point O by a thread of length ℓ moves along a horizontal circle with a constant angular velocity ω .



- (A) angular momentum is constant about O
 (B) angular momentum is constant about C
 (C) vertical component of angular momentum about O is constant
 (D) Magnitude of angular momentum about O is constant
6. Two gases have the same initial pressure, volume and temperature. They expand to the same final volume, one adiabatically and the other isothermally
- (A) The final temperature is greater for the isothermal process
 (B) The final pressure is greater for the isothermal process
 (C) The work done by the gas is greater for the isothermal process
 (D) All the above options are incorrect

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
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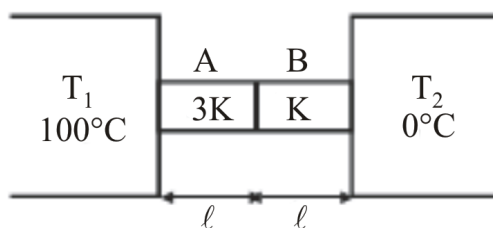
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Two rods A and B of same cross-sectional area A and length ℓ connected in series between a source ($T_1 = 100^\circ\text{C}$) and a sink ($T_2 = 0^\circ\text{C}$) as shown in figure. The rod is laterally insulated.



7. The ratio of thermal resistance of the rod is
- (A) $\frac{R_A}{R_B} = \frac{1}{3}$ (B) $\frac{R_A}{R_B} = 3$ (C) $\frac{R_A}{R_B} = \frac{3}{4}$ (D) $\frac{R_A}{R_B} = \frac{4}{3}$
8. If T_A and T_B are the temperature drop across the rod A and B then
- (A) $\frac{T_A}{T_B} = \frac{1}{3}$ (B) $\frac{T_A}{T_B} = \frac{3}{1}$ (C) $\frac{T_A}{T_B} = \frac{3}{4}$ (D) $\frac{T_A}{T_B} = \frac{4}{3}$

Space for Rough Work

Paragraph for Questions 9 and 10

Consider the potential energy given by

$$U(x) = -\frac{x^4}{4} - \frac{x^3}{3} + x^2,$$

where the coefficients of the various powers of x have units such that U has units of joules.

9. Find the equilibrium points and classify them.
- (A) $x = 0$ stable (B) $x = 1$ (stable) (C) $x = 2$ (stable) (D) $x = 0$ neutral
10. If total energy is 16 J and mass of particle is 2 kg which of the following option is incorrect
- (A) Particle will present at $x = 1$
- (B) at origin $v = 4$ m/s
- (C) Acceleration of particle at $x = 0$ will be 4m/s^2
- (D) Momentum of particle at $x = 0$ will be 8 m/s

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A shell is fired vertically upward with a speed of 60 m/s. When it is at maximum height, it explodes into large number of fragments. Assume that the fragments fly in every possible direction and all of them have same initial speed of 25 m/s.

1. The speed (in m/s) of the center of mass of the system after 1 sec of explosion will be :-
2. The minimum time taken (in sec) by any one fragment to reach at ground will be :-

Paragraph for Questions 3 and 4

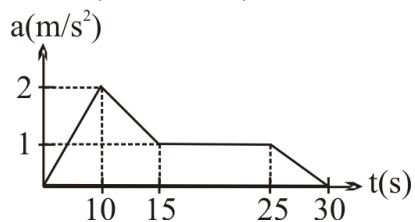
The energy radiated by a black body at 2300 K is found to have the maximum at a wavelength 1260 nm, its emissive power being 8000 Wm^{-2} . When the body is cooled to a temperature T K, the emissive power is found to decrease to 500 Wm^{-2} . Find :

3. The wavelength (in nm) at which intensity of emission is maximum at the temperature T.
4. The temperature T in K:

Space for Rough Work

Paragraph for Questions 5 and 6

In figure shown, the graph shows the variation of a acceleration a acting on a body of mass 10 kg (in gravity free space), with time t . The velocity of the body at $t = 0$ is zero.



5. The average acceleration of the body from $t = 0$ to $t = 15$ s is
6. If kinetic energy of the particle at $t = 10$ sec is k and magnitude of momentum of particle at this time is p , find ratio of p/k .

Space for Rough Work

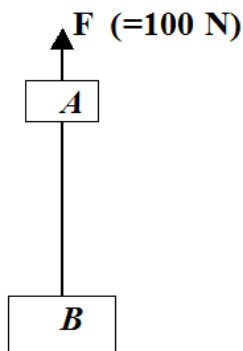
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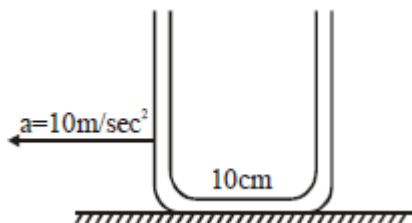
Zero Marks : 0 In all other cases

7. A car is travelling with a speed of 90 m/s due east. An observer is present at 5 m south of the line of motion. Find the angular velocity (in rad/sec) of car relative to the observer when car is located northeast of him.
8. Consider the shown arrangement where the blocks A and B connected by means of a uniform string is being moved vertically up by the force F. Each block weighs 2 kg while the mass of string is 1000 gm. The tension (in N) at bottom of the string equals. ($g = 10 \text{ m/s}^2$)



Space for Rough Work

9. The horizontal part of the U-shaped tube shown in the figure is 10 cm, and its vertical branches are long enough. The cross section of the tube is 1 cm^2 . First, 20 cm^3 water is poured into the tube, then 20 cm^3 oil is poured into the left branch. (density of water = 1000 kg/m^3 , density of oil = 800 kg/m^3). The whole system is accelerated towards the left, at an acceleration of $a = 10 \text{ m/sec}^2$. Determine the difference between the level of water and the level of oil (in cm).



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
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1. For the gas phase reaction $\text{C}_2\text{H}_4 + \text{H}_2 \rightleftharpoons \text{C}_2\text{H}_6$ $\Delta H = -136.8 \text{ kJ mol}^{-1}$ carried out in a vessel, the equilibrium concentration of C_2H_4 can be increased by

(A) Increasing the temperature

(C) Removing some H_2

(B) Decreasing the pressure

(D) Adding some C_2H_6
2. Select the correct statement(s).

(A) All electromagnetic radiation travel with speed of light in vacuum.

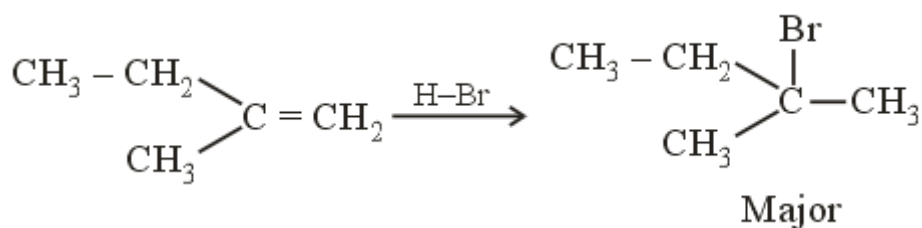
(B) Energy of photon of UV light is lower than that of yellow light.

(C) Atomic spectrum is discontinuous spectrum

(D) The total energy of an electron in H-like atom is greater than zero.

Space for Rough Work

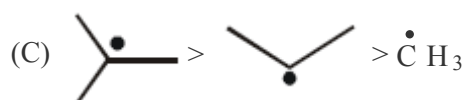
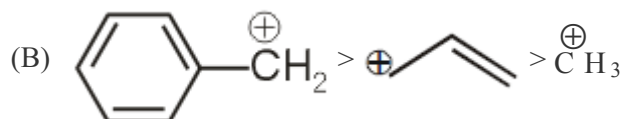
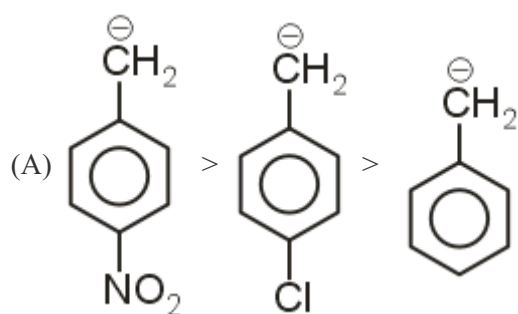
3. Which of the following set of quantum number is/are not possible for Ac?
- (A) $n = 3, l = 3, m = 0, s = +1/2$ (B) $n = 6, l = 2, m = -1, s = -1/2$
 (C) $n = 6, l = 3, m = -2, s = +1/2$ (D) $n = 7, l = 0, m = 0, s = -1/2$
4. In which of the following compound(s), atleast three X–O–X linkage are present?
 (Where X = S, B or P)
- (A) S_3O_9 (B) $H_2S_3O_6$
 (C) $Na_2B_4O_7 \cdot 10H_2O$ (D) $H_3P_3O_{10}$
5. Select correct statement(s) regarding given reaction



- (A) $\text{Br}^\ominus$ will attack as nucleophilic after carbocation formation
 (B) Major product formed according to Markovnikov's rule.
 (C) Carbocation is formed as an intermediate
 (D) It is substitution reaction

Space for Rough Work

6. Identify correct stability order for following intermediats / ions ?



Space for Rough Work

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Paragraph for Questions 7 and 8



In above diagram A has five valence electrons and B, C and D have six valence electrons each. According to the Lewis dot structure, A has **x** formal charge, B has **y** and C has **z** formal charge. The above mentioned compound / ion follows Lewis octet theory. Give the answer of following question.

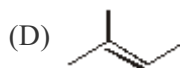
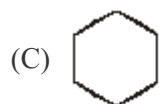
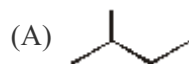
7. Identify the formal charge of A, B and C respectively.
- (A) -2, 0, +1 (B) +1, 0, +1 (C) -2, 0, +2 (D) +1, 0, -1
8. Choose the incorrect options?
- (A) B, C, D belong to chalcogen family.
- (B) Compound/ion has $p\pi-p\pi$ bond
- (C) Central atom A has sp^2 hybridisation
- (D) The sum of formal charge is zero in the molecule/ion.

Space for Rough Work

Paragraph for Questions 9 and 10



9. Which one of the following is an isomer of all of above



10. How many compounds in paragraph will have word root 'Pent' in their IUPAC name

(A) One

(B) Two

(C) Three

(D) Four

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

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Paragraph for Questions 1 and 2

The VSEPR model considers double and triple bonds to have slightly greater repulsive effect than single bonds because of the repulsive effect of π -electrons. However the lone pair creates the maximum repulsive effect.

1. Total number of planes where maximum number of atoms lying in one plane in IF_5 molecule :
2. In which of the following molecular shape, $d_{x^2-y^2}$ -orbital is not involved in bonding (with respect to central atom)?
 BF_3 , CH_4 , CO_2 , PCl_5 , PH_4^+ , SF_6

Space for Rough Work

Paragraph for Questions 3 and 4

In permanganate titrations, potassium permanganate is used as an oxidizing agent in acidic medium. The medium is maintained acidic by the use of dilute sulphuric acid. Potassium permanganate acts as self indicator. The potential equation, when potassium permanganate acts as an oxidizing agent is

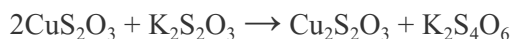
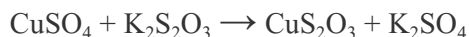


Before the end point, the solution remains colourless but after the equivalence point only one extra drop of KMnO_4 solution imparts pink colour, i.e. appearance of pink colour indicates end point. These titrations are used for estimation of ferrous salts, oxalic acid, oxalates, hydrogen peroxide, As_2O_3 etc.

3. In order to prepare one litre of 1N KMnO_4 , how many grams of KMnO_4 is required if solution is to be used in acid medium for oxidation. [$M_{\text{KMnO}_4} = 158$]
4. For decolourization of 1 mol of KMnO_4 , the number of moles of $\text{H}_2\text{C}_2\text{O}_4$ required is :-

Paragraph for Questions 5 and 6

570 g of potassium thiosulphate ($\text{K}_2\text{S}_2\text{O}_3$) reacts with copper sulphate to form cupric thiosulphate which is reduced by remaining potassium thiosulphate to give cuprous compound which reacts with the remaining potassium thiosulphate to form a complex compound potassium cuprothiosulphate ($\text{K}_4[\text{Cu}_6(\text{S}_2\text{O}_3)_5]$).



In this process, 0.2 mole of potassium cuprothiosulphate is formed. (O = 16, K = 39, S = 32)

5. Total moles of potassium thiosulphate reacted in the reactions is -
6. If instead of given amount of potassium thiosulphate, 2 moles of potassium thiosulphate along with 3 moles of CuSO_4 were taken initially, then moles of potassium cuprothiosulphate formed is -

Space for Rough Work

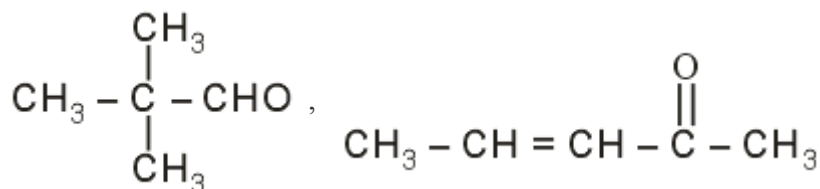
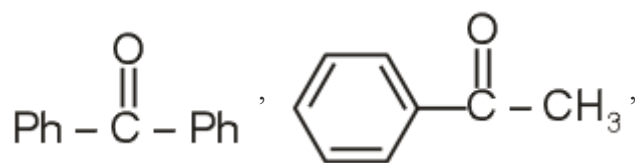
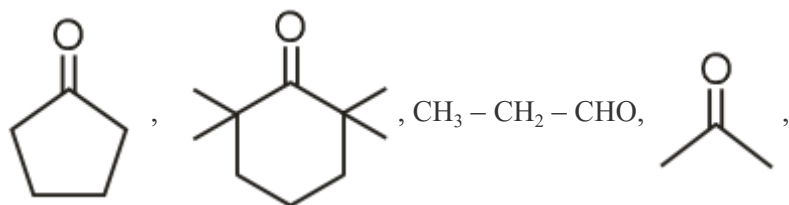
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Zero Marks : 0 In all other cases

- Calculate minimum amount in gram of NaOH required to neutralize completely an acid mixture of 0.1M, 100 ml of H_2SO_4 and 0.2M, 400 ml of HCl.
- Total number of ring structure formed with molecular formula C_5H_{10} is :
- Find number of compound which can show Tautomerism :



Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. If the roots of the equation $x^2 - 4x - \log_2 a = 0$ are real, then 'a' belongs to -
 (A) $a \geq \frac{1}{32}$ (B) $a \geq 0$ (C) $a \geq \frac{1}{16}$ (D) $a \geq \frac{1}{4}$
 2. A line passing through the point (3, -2) and inclined at 60° to the line $\sqrt{3}x + y = 1$, can be -
 (A) $x + 2 = 0$ (B) $y + 2 = 0$
 (C) $\sqrt{3}x + y = 2 + 3\sqrt{3}$ (D) $\sqrt{3}x - y = 2 + 3\sqrt{3}$
 3. For a positive integer n, if the expansion of $\left(x^7 - \frac{3}{x}\right)^n$ has a term independent of x then n can be
 (A) 8 (B) 12 (C) 16 (D) 24
-

Space for Rough Work

4. If 'n' different objects are given in a row, then the number of ways of selecting three of these objects so that no two adjacent objects are selected is/are -
 (A) ${}^{n-2}C_3$ (B) ${}^{n-2}C_{n-5}$ (C) ${}^{n-3}C_2$ (D) ${}^{n-3}C_{n-5}$
5. For the word "PLATONIC" which of the following is/are correct ?
 (A) Number of four letter words with no letter being repeated formed is 336.
 (B) Number of four letter words with no letter being repeated formed is 1680.
 (C) Number of four letter words with no letter being repeated formed such that the letter P always appear in every word is 840.
 (D) Number of four letter words with no letter being repeated formed such that the letter P always appear in every word is 210.
6. If the sum of n terms of an arithmetic progression is given by $S_n = 2n^2 + 3n$, having a_n and d as their n^{th} term and common difference respectively, then
 (A) $d = 4$ (B) $d = 3$ (C) $a_2 = 9$ (D) $a_2 = 5$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Tangent are drawn from the origin "O" to the circle $x^2 + y^2 - 8x - 6y + 16 = 0$ touches it at the points A and B respectively.

On the basis of above information, answer the following :

7. If $\angle AOB = \alpha$, then $\sin \alpha$ is equal to :-
- (A) $\frac{3}{5}$ (B) $\frac{4}{5}$ (C) $\frac{12}{25}$ (D) $\frac{24}{25}$
8. Equation of circle, circumscribing the triangle OAB is :-
- (A) $x^2 + y^2 - 4x - 3y = 0$ (B) $x^2 + y^2 - 3x - 4y = 0$
- (C) $x^2 + y^2 + 4x + 3y = 0$ (D) $x^2 + y^2 + 3x + 4y = 0$

Paragraph for Questions 9 and 10

For an ellipse $\frac{x^2}{32} + \frac{y^2}{18} = 1$, line $L = 4x - 3y = 7$ is normal at point P. Then

9. Area enclosed by ellipse is :
- (A) 6π (B) 12π (C) 24π (D) 36π
10. Coordinate of point P is :
- (A) $(1, -1)$ (B) $(2, 1/3)$ (C) $(3, 4)$ (D) $(4, 3)$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Consider 3 alike english books, 5 alike hindi books and 3 different science books.

1. How many different selections can be made so that each selection consists of exactly one book on each subject ?
2. How many different selections can be made so that each selection consist of taking atleast one book ?

Paragraph for Questions 3 and 4

If all the letters of word "CHECKUARY" are used

3. Number of words can be formed in which no two vowels are together are λ , then the value of $\frac{\lambda}{100}$ is
4. Number of words can be formed without changing the relative order of vowels and consonents is k, then the value of $\frac{k}{10}$ is

Space for Rough Work

Paragraph for Questions 5 and 6

Consider an AP 65, 62, 59, 56,

5. Maximum value of sum of A.P. is -
6. The value of first negative term of the series is k , then the value of $\left| \frac{k}{4} \right|$ is -
-

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. Number of terms free from radical sign in the expansion of $(\sqrt{5} + \sqrt[4]{11})^{100}$ is
8. The length of the chord of the parabola $y^2 = x$ which is bisected at the point (2, 1) is $k\sqrt{5}$. Then value of k is
9. If the number of ways in which word "KOLAVARIDI" can be arranged is n_1 and n_2 is the number of ways such that all vowels are together, then the value of sum of digits of $\left(\frac{n_1}{n_2}\right)$ is

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work